Lecture 26

Higher homotopy groups and topological defects

Objectives. *A loop detects whether a line defect can be removed while the surrounding medium stays ordered. A sphere provides the corresponding test for a point defect. We extend the fundamental group to maps from higher-dimensional spheres. We then distinguish based charges from the free homotopy classes that describe local defects. This distinction identifies opposite point charges for unoriented axes, but preserves their sign for oriented magnetic directions.*

By the end of this lecture you will be able to:

- Prove the group laws and commutativity of higher homotopy groups.
- Use the cited sphere groups and fibration sequence in the stated cases.
- Choose the linking circle for a line defect and sphere for a point defect.
- Compute vortex, nematic and sphere-valued charges with the required basepoint qualification.

26.1 Maps from cubes and higher homotopy groups

Let X be a space with basepoint $x_0 \in X$. For $n \geq 1$, put $I^n = [0,1]^n$. The *higher homotopy group* $\pi_n(X, x_0)$ is initially the set of continuous maps $f\colon I^n \to X$ sending the entire boundary to x_0, modulo homotopies that keep that boundary at x_0. Collapsing the boundary identifies $I^n/\partial I^n$ with S^n. Thus the same classes are based maps from a sphere. For $n = 1$ this is Lecture 3's fundamental group. The symbol $\pi_0(X)$ instead denotes the set of path components. It is not generally a group.

Write $t' = (t^2, \dots, t^n)$. Concatenation in the first coordinate is given by

$$(f+g)(t^1, t') = \begin{cases} f(2t^1, t'), & 0 \leq t^1 \leq 1/2, \\ g(2t^1 - 1, t'), & 1/2 \leq t^1 \leq 1. \end{cases}$$

Both seam values are x_0, so pasting gives a continuous map relative to the boundary. For $n = 1$, this operation traverses f first, in the course's path convention.

Proposition 26.1 (The group structure · *proved*)**.** *Concatenation makes $\pi_n(X, x_0)$ a group for every $n \geq 1$. Its identity is the constant map and its inverse is represented by $f^-(t^1, t') = f(1 - t^1, t')$. For $n \geq 2$ the group is abelian.*

Proof. Pasting homotopies on the two halves shows that concatenation is well-defined on classes. The two bracketings of three maps occupy first-coordinate slabs of lengths $(1/4, 1/4, 1/2)$ and $(1/2, 1/4, 1/4)$. For a homotopy parameter $u \in [0,1]$, use slab lengths

$((1+u)/4, 1/4, (2-u)/4)$ and rescale the first coordinate on each slab affinely to the unit interval. All lengths are positive. Adjacent maps agree at their boundary value x_0. This continuous reparametrization proves associativity.

A constant first slab is a pause before f. Shrink its length a continuously from $1/2$ to zero, using the value x_0 for $t^1 \le a$ and $f((t^1-a)/(1-a), t')$ for $t^1 \ge a$. This is a homotopy relative to the cube boundary and proves the left unit law. Shrinking a final pause proves the right unit law. For $f + f^-$ the explicit retracing contraction is

$$K_u(t^1, t') = f\big((1-u)\min\{2t^1, 2-2t^1\}, t'\big).$$

At $u = 0$ it is the concatenation and at $u = 1$ it is the constant boundary value. For $f^- + f$, replace the first argument by $1-(1-u)\min\{2t^1, 2-2t^1\}$, ending at the other boundary face. This proves both inverse laws.

For $n \ge 2$, let $\star$ concatenate in the second coordinate. The same proof gives it the same identity, denoted 0 at the class level. Pasting four maps into quadrants gives the exact interchange identity shown in the figure:

$$(a+b)\star(c+d) = (a\star c)+(b\star d).$$

Here $a, b, c, d \in \pi_n(X, x_0)$. Its two specializations give

$$a+b = (a\star 0)+(0\star b) = (a+0)\star(0+b) = a\star b, \qquad a+b = (0\star a)+(b\star 0) = (0+b)\star(a+0) = b\star a.$$

The operations therefore agree and are commutative. ∎

The figure uses only the first two coordinates. The other coordinates stay fixed.

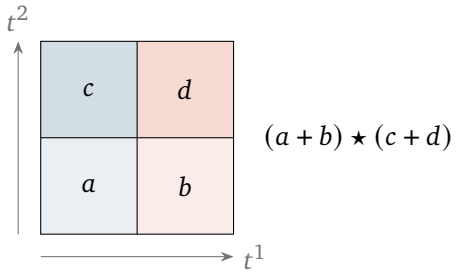


Figure 26.1: The four cube maps have the same placement whether one first joins horizontal pairs or first joins vertical pairs. All seam faces map to the basepoint.

For example, based maps $S^2 \to S^2$ add by joining two sphere maps at their basepoint. The next section identifies this group law with addition of degrees.

26.2 Sphere groups and the Hopf map

The degree from Lecture 14 is invariant under homotopy. The converse for sphere maps is an additional theorem: equal degrees imply based homotopy. We cite it here rather than infer it from the integral formula alone.

Theorem 26.2 (Homotopy groups of spheres · *cited*). *For every $n \ge 1$, the degree map gives $\pi_n(S^n) \cong \mathbb{Z}$. For $1 \le k < n$, the group $\pi_k(S^n)$ vanishes. The group $\pi_3(S^2)$ is infinite cyclic and is generated by the Hopf map $S^3 \to S^2$ of Lecture 20.*

The precise sources are Hatcher, Corollary 4.9, p. 349, Corollary 4.25, p. 361, and Example 4.51, p. 380. Hatcher's homological degree is the integer multiplier d on $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$. This definition is given in §2.2, p. 134. This multiplier does not depend on a choice of generator. For smooth maps, Lecture 14's cited de Rham identification intertwines this action with pullback of forms: integrating a form over a composed simplex equals integrating its pullback over that simplex. Thus the homological degree agrees with the integral degree of Theorems 14.2 and 14.6. To verify this, let c be the period of a normalized volume form ν on a generator $[S]$. Its de Rham class is nonzero since $\int_{S^n} \nu = 1$, so injectivity of the pairing gives $c \neq 0$. The top-cohomology argument in Lecture 14 gives $[f^*\nu] = (\deg f)[\nu]$. Naturality and the vanishing of exact-form periods therefore give

$$dc = \int_{f_*[S]} \nu = \int_{[S]} f^*\nu = (\deg f)c.$$

Cancellation of c identifies the two degrees. For continuous maps, Hatcher's definition uses homology directly. Thus the identity sphere map has charge +1 and generates $\pi_n(S^n)$. Maps from a smaller-dimensional sphere cannot carry such a charge. The Hopf result also shows that higher homotopy groups need not vanish when the domain sphere is larger.

We will also use the covering-space result in Hatcher, Proposition 4.1, p. 342: a covering induces an isomorphism on π_n for $n \geq 2$. Sphere maps and their homotopies lift, which explains this result. The contractible cover $\mathbb{R} \to S^1$ gives $\pi_n(S^1) = 0$ for $n \geq 2$. The cover $S^2 \to \mathbb{R}\mathrm{P}^2$ gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$ after choosing a lifted basepoint. That choice will matter when the basepoint is allowed to move.

26.3 The fibration sequence and two consequences

A *fibration* is a continuous map $p\colon E \to B$ with the homotopy lifting property: for every space Y, a homotopy $Y \times [0,1] \to B$ with a chosen lift at its initial time has a continuous lift throughout that restricts to the chosen lift at time zero. For the theorem below it suffices to require this property for disks of every dimension. Fibre bundles have this property for disks by Hatcher's Proposition 4.48, p. 379. Choose a basepoint $b_0 \in B$ and a point e_0 in its fibre $F = p^{-1}(b_0)$. All homotopy groups below use these compatible basepoints.

Theorem 26.3 (The fibration sequence · *cited*)**.** *For a fibration over a path-connected base, the homotopy groups have a long exact sequence. Its relevant segment is*

$$\cdots \longrightarrow \pi_2(F) \longrightarrow \pi_2(E) \longrightarrow \pi_2(B) \xrightarrow{\partial_{\mathrm{fib}}} \pi_1(F) \longrightarrow \pi_1(E) \longrightarrow \pi_1(B) \longrightarrow \cdots .$$

Exactness means that the image of each arrow is the kernel of the next.

This is Hatcher's Theorem 4.41, p. 376. The symbol ∂_{fib} names the connecting homomorphism, distinct from a geometric boundary. Represent a based sphere by a square with its boundary collapsed. Lifting this square leaves a boundary loop in the fibre. Its construction and the general exactness proof are cited.

For the Hopf bundle, $F = S^1$, $E = S^3$ and $B = S^2$. The cited sphere groups give $\pi_2(S^3) = 0$ and $\pi_1(S^3) = 0$. The figure isolates the resulting segment.

$$0 = \pi_2(S^3) \longrightarrow \pi_2(S^2) \xrightarrow[\cong]{\partial_{\mathrm{fib}}} \pi_1(S^1) \longrightarrow \pi_1(S^3) = 0$$

Figure 26.2: Exactness makes the Hopf connecting map an isomorphism: its kernel is zero and its image is the entire fibre-loop group.

Hence $\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$. This is a consequence of the cited sequence, not a second proof of that sequence.

More generally, let G be a simply connected Lie group with $\pi_2(G) = 0$ and let H be a closed Lie subgroup. Lecture 20 gives the principal bundle $H \to G \to G/H$. Exactness gives $\pi_2(G/H) \cong \pi_1(H)$ because both adjacent groups from G vanish. Here the basepoint of H is its identity, so its loops lie in its identity component. The assumption $\pi_2(G) = 0$ is explicit. Simple connectedness by itself is not this assumption.

26.4 Order parameters and moving basepoints

An *order parameter* records the local ordered state of a medium. Its allowed ordered values form the *order-parameter space* $\mathcal{M}$. When a symmetry group G acts transitively on the uniform ordered states, Lecture 8 identifies this space with G/H, where H fixes a reference state. The relevant descriptions in this lecture are a superfluid phase in S^1, a unit ferromagnetic direction in S^2, and a nematic director in $\mathbb{R}\mathrm{P}^2$. A director is an unoriented unit axis: $\mathbf{n}$ and $-\mathbf{n}$ describe the same state. The word *texture* means a continuous spatial configuration of the order parameter.

For an ordinary based charge, a chosen point of the domain sphere must have a prescribed image x_0. A deformation of the physical texture can move that image. We therefore need the effect of moving the basepoint. Let $\gamma\colon [0,1] \to \mathcal{M}$ run from x_0 to x_1. Rescale the domain of a cube map based at x_1 onto a smaller concentric cube. On each radial segment of the surrounding shell, insert γ, running from x_0 at the outer boundary to x_1 at the inner boundary. The figure depicts this construction.

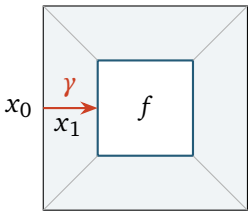


Figure 26.3: The inner cube carries a map based at x_1. The shell changes its basepoint to x_0 by the path γ. The outer boundary maps constantly to x_0.

Proposition 26.4 (The basepoint action · *sketched*)**.** *The shell construction gives an isomorphism $\beta_\gamma : \pi_n(\mathcal{M}, x_1) \to \pi_n(\mathcal{M}, x_0)$. For loops at x_0, it gives an action of $\pi_1(\mathcal{M}, x_0)$ on $\pi_n(\mathcal{M}, x_0)$. For $n = 1$ this is conjugation.*

Sketch. A homotopy of the path or inner map gives a homotopy of the shell map. Reversing the path gives the inverse isomorphism, since the retraced shell contracts. Two nested shells concatenate their paths. Consequently $\beta_{\gamma\eta} = \beta_\gamma \beta_\eta$ in the first-path-first convention. Joining inner cube maps shows that the induced maps preserve the group operation. For $n = 1$ the shell map is the path $\gamma f \gamma^{-1}$, as in Lecture 3. These constructions give the asserted action. They are described in Hatcher, pp. 341–342. ∎

Even though π_n is abelian for $n \geq 2$, this external action can be nontrivial. Commutativity of its own group law does not remove the basepoint issue.

26.5 Local defects and linking spheres

A *defect core* is a region where the chosen ordered description fails. Outside the cores the order parameter is a continuous map into $\mathcal{M}$. A small circle surrounding a line in three-dimensional space is a *linking circle*. A small sphere surrounding an isolated point is its *linking sphere*. The figure separates these two domains of the order-parameter map.

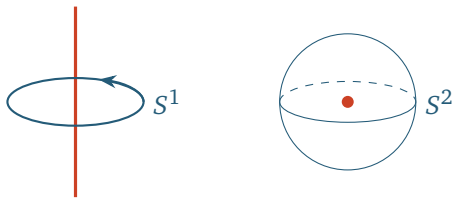


Figure 26.4: A circle links a line core. A sphere encloses a point core. The order parameter is required to remain in $\mathcal{M}$ on the linking domain throughout a deformation.

For a core of codimension c, the local transverse boundary is S^{c-1}. This explains which homotopy group occurs. The physical sphere's orientation is fixed. Reversing it can reverse an integer charge. A *topological defect* has a nontrivial linking homotopy class that obstructs removal of its core within $\mathcal{M}$.

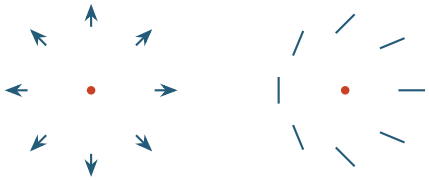
Theorem 26.5 (Local defect classes · *sketched*)**.** *Let $\mathcal{M}$ be a path-connected order-parameter manifold. Classify restrictions to a linking domain by continuous homotopy through maps into $\mathcal{M}$. Without a fixed image basepoint, line defects in three dimensions have the conjugacy classes of $\pi_1(\mathcal{M})$, and point defects have the orbits of $\pi_2(\mathcal{M})$ under the $\pi_1(\mathcal{M})$ action. A two-dimensional texture with a fixed uniform value at spatial infinity has a based class in $\pi_2(\mathcal{M})$.*

Sketch. The first assertions concern free homotopy classes of maps from S^1 and S^2. To convert a free homotopy to a based one, follow the moving image of the chosen domain point. Its path is precisely the shell change of basepoint. Thus representatives of two based classes are freely homotopic exactly when this action identifies them. For loops the action is conjugation, recovering Lecture 3's full loop result. For spheres this is its higher-dimensional version (Mermin, §VII.D, p. 631). For the texture, adding the point at infinity compactifies the oriented plane to S^2. The fixed limiting value supplies its target basepoint. This gives the stated based class. ∎

If the value at infinity is also allowed to move, the texture likewise needs action orbits. A nonzero linking class obstructs extension across the core within $\mathcal{M}$: an extension to the transverse disk would contract the boundary map radially. A zero class admits such an extension because a representative map is null-homotopic. This is a topological obstruction, not a claim about the energy or time needed to remove the core. The local boundary classes do not classify knotting or all configurations with several defects.

26.6 Vortices, nematics and magnetic textures

The figure compares an oriented phase field with an unoriented director field on a linking circle.



$e^{i\phi}$ $\qquad$ $[\cos(\phi/2) : \sin(\phi/2) : 0]$

Figure 26.5: The phase turns once on a superfluid linking circle. The nematic axis turns through a half-turn and already returns to the same unoriented axis.

Example 26.6 (Three order-parameter spaces)**.** Determine the local line and point classes for a superfluid, a uniaxial nematic and a Heisenberg ferromagnet.

1. A superfluid has a phase $e^{i\alpha} \in S^1$. On an oriented linking circle a vortex of winding $n \in \mathbb{Z}$ is represented by $\phi \mapsto e^{in\phi}$ for $0 \le \phi \le 2\pi$. Lecture 3 gives $\pi_1(S^1) = \mathbb{Z}$. This group is abelian, so its conjugacy classes are its individual integers. Based joining adds windings, which is also the local fusion rule for these free classes. Two vortices of winding +1 have total winding 2 and cannot cancel within S^1. Opposite windings have total zero and have no such obstruction. The group $\pi_2(S^1)$ vanishes, so there is no corresponding point charge.
2. A uniaxial nematic has $\mathcal{M} = \mathbb{R}P^2$. Lecture 4 gives $\pi_1(\mathbb{R}P^2) = \mathbb{Z}_2$. The nontrivial line defect is called a *disclination*. A representative director is $[\cos(\phi/2) : \sin(\phi/2) : 0]$. Its vector lift starts at a unit vector and ends at its negative. The director nevertheless closes, since antipodal vectors give the same axis. Joining two such loops gives a closed full-turn vector loop on S^2. That loop contracts because S^2 is simply connected. Thus the two nontrivial disclinations have total class zero and can cancel topologically. The chosen sphere cover gives $\pi_2(\mathbb{R}P^2) \cong \mathbb{Z}$. Changing the lifted basepoint around the nontrivial director loop applies the antipodal deck map. That map has degree -1 on S^2, so its action sends n to $-n$. Hence free point-defect classes identify opposite integers. They can be labelled by $|n|$, including zero. There is no single signed addition law on these orbits without the joining-path data. This is the nematic qualification in Mermin, p. 631.
3. A Heisenberg ferromagnet has $\mathcal{M} = S^2$. The fundamental group of this target is zero, so there is no local line obstruction of this type. Its point classes are the signed integers $\pi_2(S^2) = \mathbb{Z}$, with no nontrivial basepoint action. The *hedgehog* is the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ outside the origin. Its restriction to an outward linking sphere is the identity direction map and has degree +1.
   A *skyrmion texture* is a nonsingular plane texture with a nonzero sphere degree and a fixed uniform value at infinity. For $a > 0$, an explicit degree-one example is

$$\mathbf{n}(x, y) = \frac{(2ax, 2ay, a^2 - x^2 - y^2)}{a^2 + x^2 + y^2}, \qquad \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{4a^2}{(a^2 + x^2 + y^2)^2}.$$

   Its norm is one. It tends to the south pole at infinity. For the density calculation, write the field's numerator as $\mathbf{d}$ and denominator as $D = a^2 + x^2 + y^2$. Differentiating

$\mathbf{n} = \mathbf{d}/D$ gives

$$\mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{\mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d})}{D^3}, \qquad \mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d}) = 4a^2 D.$$

The terms proportional to $\mathbf{d}$ in either derivative drop out of the determinant. In inversion coordinates $X = x/(x^2 + y^2)$ and $Y = y/(x^2 + y^2)$, the same texture is

$$\mathbf{n} = \frac{(2aX, 2aY, a^2(X^2 + Y^2) - 1)}{1 + a^2(X^2 + Y^2)}.$$

It extends smoothly to $X = Y = 0$, the point at infinity. Its degree is

$$\frac{1}{4\pi} \int_{\mathbb{R}^2} \frac{4a^2 \mathrm{d}x\mathrm{d}y}{(a^2 + x^2 + y^2)^2} = \int_0^\infty \frac{2a^2 r \mathrm{d}r}{(a^2 + r^2)^2} = 1.$$

This is a texture, with no singular core in the plane. The degree labels its topology. No energy-minimizing field equation has been imposed.

Self-check. An ordered medium has $\mathcal{M} = \mathrm{SO}(3)$. Find its local line classes and the result of joining two nontrivial lines.

Answer. Lecture 4 gives $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. It is abelian, so there are two free line classes. The sum of two nontrivial classes is zero, removing this local topological obstruction. This does not assert that a particular physical evolution makes the defects annihilate.

Summary. The groups π_n classify based sphere maps and are abelian for $n \geq 2$. The cited sphere results and fibration sequence compute the groups used here. Local free defect classes still require conjugacy for loops and the fundamental-group action for spheres. Superfluid vortices retain signed integer winding. Nematic disclinations have $\mathbb{Z}_2$ charge. Free nematic point charges identify n with $-n$. Sphere-valued hedgehogs and based plane textures have signed degree. Lecture 27 adds boundary conditions and energy to this topological classification.

Reading. Hatcher, §4.1, pp. 340–343 and 349; §4.2, Corollary 4.25, p. 361, Theorem 4.41, p. 376, Proposition 4.48, p. 379, and Example 4.51, p. 380. Mermin, *The topological theory of defects in ordered media*, §VII.D–E, pp. 630–632. Nakahara, §4.8–4.9.